

Programming Lab 6 - Functions and Program Decomposition

Case Problem Labs

In this lab, you will refactor your existing case problem by organizing the program into multiple functions. Rather than placing all program logic in a single block of code, you will create functions that perform specific tasks and work together to complete the program.

Your solution should maintain the business rules and calculations from previous labs while demonstrating effective use of functions.

Learning Objectives

By completing this lab, students will:

- Create and call user-defined functions.
- Use parameters to pass information into functions.
- Return values from functions.
- Decompose a larger program into smaller, reusable components.
- Improve program organization and readability.

Technical Requirements (All Case Problems)

Your program must:

- Contain a minimum of **five user-defined functions**.
- Include a **main()** function.
- Use parameters appropriately.
- Return values appropriately.
- Use meaningful variable and function names.
- Include comments describing the purpose of each function.
- Maintain all business rules from previous labs.
- **Use exception handling for all numeric input. The program should prevent crashes by handling invalid user input and prompting the user to enter a valid value.**

Important Note

Programming labs often require multiple attempts and revisions. Debugging and refining your program are normal parts of learning to program.

★ Challenge Extensions (Optional)

Challenge Extensions are optional enhancements for students who would like additional practice or a more advanced programming challenge. Students should focus on completing the required functionality first before attempting extensions.

Cost Is No Object

The Barking Lot

Fountain View Hall

Cost Is No Object

Scenario

Modify your existing car rental program to use functions.

Required Functions

Students must create at least five functions, including:

- `get_customer_information()`
 - `get_rental_information()`
 - `calculate_rental_charge()`
 - `calculate_discount()`
 - `display_reservation_summary()`
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Program Requirements

- Gather customer information.
 - Gather rental information.
 - Calculate rental charges.
 - Apply all existing business rules.
 - Display a formatted rental summary.
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Function Requirements

At least:

- Two functions must accept parameters.
 - Three functions must return values.
 - A `main()` function must control overall program execution.
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★ Challenge Extensions (Optional)

Add one or more of the following:

- a function to validate user input.
 - a function to calculate taxes.
 - a function to determine loyalty discounts.
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Submission Requirements

Submit:

- `.py` file
 - Test cases
 - Code walk-through video (3–4 minutes) demonstrating program runs and explaining how functions were used to organize the solution.
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The Barking Lot

Scenario

Modify your existing dog boarding program to use functions.

Required Functions

Students must create at least five functions, including:

- `get_owner_information()`
 - `get_dog_information()`
 - `calculate_boarding_charge()`
 - `calculate_additional_services()`
 - `display_boarding_summary()`
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Program Requirements

The program must:

- Gather owner information.
 - Gather dog information.
 - Calculate boarding charges.
 - Apply all existing business rules.
 - Display a formatted boarding summary.
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Function Requirements

At least:

- Two functions must accept parameters.
 - Three functions must return values.
 - A `main()` function must coordinate program flow.
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★ Challenge Extensions (Optional)

Add one or more of the following:

- a function for boarding validation.
 - a function for special-service calculations.
 - a function for manager approval checks.
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Submission Requirements

Submit:

- `.py` file
 - Test cases
 - Code walk-through video (3–4 minutes) demonstrating program runs and explaining how functions were used to organize the solution.
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Fountain View Hall

Scenario

Modify your existing event reservation program to use functions.

Required Functions

Students must create at least five functions, including:

- `get_client_information()`
- `get_event_information()`
- `calculate_event_charge()`
- `calculate_deposit()`
- `display_reservation_summary()`

Program Requirements

The program must:

- Gather client information.
- Gather event information.
- Calculate event charges.
- Apply all existing business rules.
- Display a formatted reservation summary.

Function Requirements

At least:

- Two functions must accept parameters.
- Three functions must return values.
- A `main()` function must coordinate program execution.

★ Challenge Extensions (Optional)

Add one or more of the following:

- a function to calculate catering costs.
- a function to validate attendance limits.
- a function to calculate promotional discounts.

Submission Requirements

Submit:

- `.py` file
- Test cases
- Code walk-through video (3–4 minutes) demonstrating program runs and explaining how functions were used to organize the solution.